PROJECT REPORT

Title: Li-Fi Based Appliances Control System Prototype

1. Abstract

This project presents a Li-Fi(Light fidelity) based appliance control system designed for operation in electromagnetic interference (EMI) restricted industrial environments. The system demonstrates wireless data transmission using visible light instead of radio frequency (RF) waves.

An Arduino Uno-based transmitter encodes ON-OFF control commands using On-Off Keying (OOK) modulation through a 1W LED source. A solar panel-based optical receiver connected to a NodeMCU decodes the transmitted signal and controls electrical loads via a dual-channel relay module.

The prototype validates that visible light communication can reliably toggle loads without generating electromagnetic interference, making it suitable for EMI-sensitive zones such as automation plants, hospitals, and high-voltage installations.

2. Problem Statement

In industrial automation sectors, certain zones restrict RF-based communication due to:

- Electromagnetic interference risks
- Safety regulations
- Signal congestion
- Security vulnerabilities

There is a need for a low-cost, EMI-free communication method to control machine loads wirelessly within short distances.

3. Proposed Solution

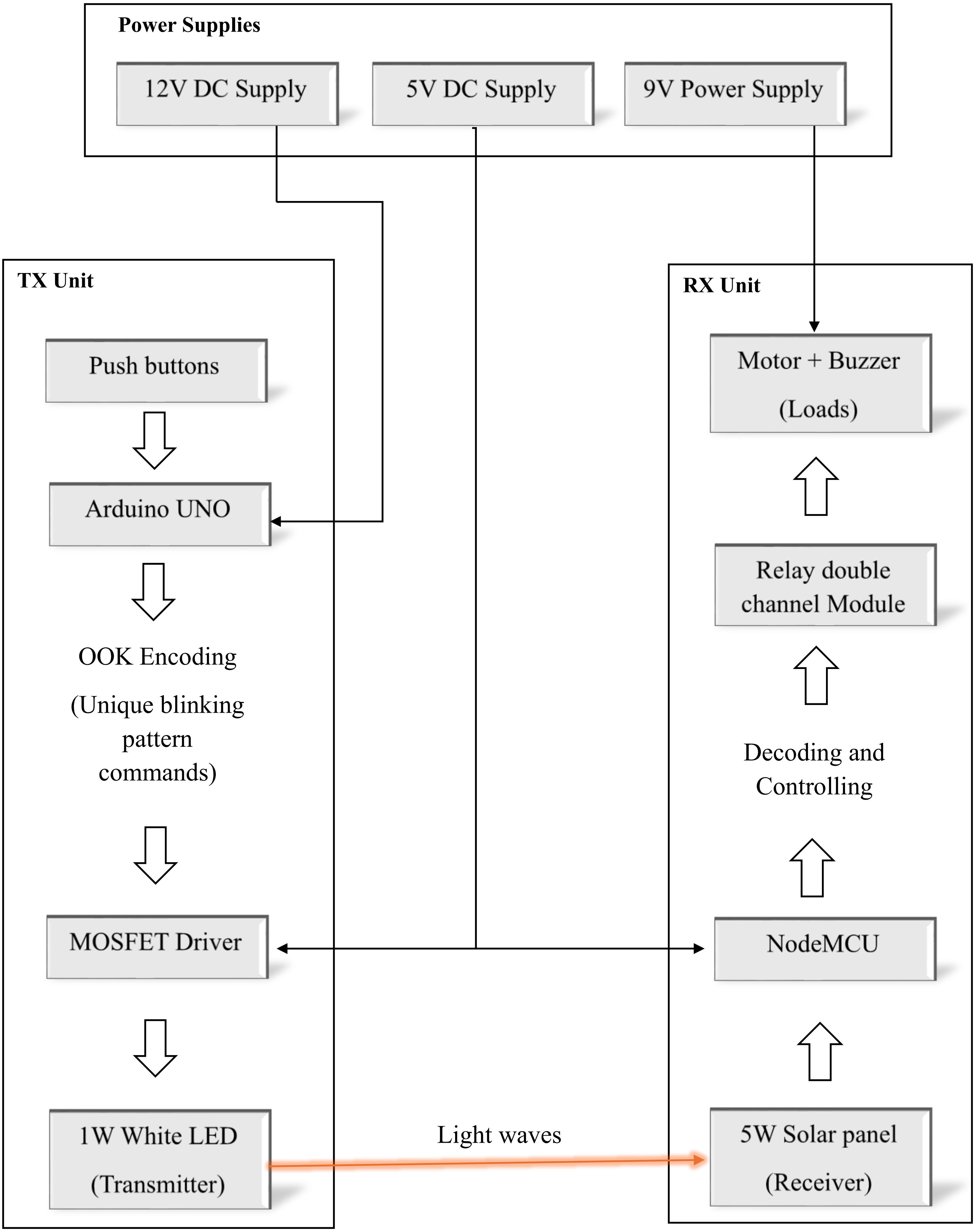
The project proposes a Li-Fi based control system that:

- Uses visible light for wireless communication
- Implements On-Off Keying modulation
- Transfers binary control commands
- Toggles loads without RF transmission

- This system eliminates electromagnetic radiation concerns and operates in unlicensed visible spectrum.

4. System Architecture:

OVERALL BLOCK DIAGRAM



4.1 Transmitter Section

Microcontroller: Arduino Uno
LED Source: 1W High-Power White LED
Driver: IRFZ44N MOSFET (LED switching)
Input Interface: Push Buttons

Function:

- Push button press generates digital command
- Arduino encodes command using OOK modulation
- MOSFET switches LED at high speed
- Binary light pulses transmitted through visible light

Each push button toggles load state (ON/OFF)

4.2 Receiver Section

Microcontroller: NodeMCU
Optical Sensor: 5W Solar Panel (used as photodetector)
Output: Dual-channel Relay Module
Loads:

- 5V DC Motor
- Passive Buzzer

Power Supply:

- 5V, 2A (Receiver + Load)
- 12V, 1A (Transmitter side supply)

Function:

- Solar panel converts light pulses into electrical signals
- NodeMCU reads analog variation
- Decodes OOK signal
- Activates relay to toggle load

Command feedback observed via Thonny IDE serial monitor:

- "Buzzer ON"
- "Buzzer OFF"

- "Fan ON"
- "Fan OFF"

5. Working Principle

The system uses On-Off Keying (OOK) modulation:

- LED ON → Logic 1
- LED OFF → Logic 0

The solar panel detects light intensity changes. These variations are interpreted as digital pulses by the NodeMCU, which then triggers corresponding relay actions.

6. Hardware Components

Transmitter:

- Arduino Uno
- 1W LED
- IRFZ44N MOSFET
- Push buttons
- 12V power supply

Receiver:

- NodeMCU
- 5W Solar panel (as photodetector)
- Dual-channel relay
- 5V DC motor
- Passive buzzer
- 5V power supply

Software:

- Arduino IDE (Transmitter programming)
- Thonny IDE (NodeMCU programming)

7. Performance Analysis

- Transmission Range: ~10 cm
- Modulation: On-Off Keying
- Communication Type: Line-of-sight
- Response: Real-time toggling

Limitations Observed:

- Limited range due to solar panel response time
- Slower response compared to photodiodes
- Sensitive to ambient light

8. Innovation & Differentiation

This project demonstrates:

- RF-free wireless control system
- EMI-safe communication method
- Use of solar panel as low-cost photodetector
- Industrial automation application

Unlike Wi-Fi or RF modules, this system:

- Does not create electromagnetic radiation interference
- Operates in visible spectrum
- Avoids spectrum congestion

9. Limitations

- Requires direct line-of-sight
- Short transmission distance
- Low data rate
- Solar panel not optimized for high-speed detection

10. Future Scope

- Replace solar panel with BPW34 photodiode for higher speed
- Increase transmission range
- Implement error detection protocol
- Apply Manchester encoding for reliable data transmission
- Integrate industrial machine communication
- Develop Li-Fi based PLC replacement system

11. Conclusion

The prototype successfully demonstrates that Li-Fi can be used for wireless appliance control in EMI-sensitive environments. While the current system is limited in range and speed, it validates the practical feasibility of visible light communication for industrial automation.

With improved optical sensors and advanced modulation techniques, the system can be scaled for real-world industrial applications.